

Abdalrhman Gamal

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LinkedIn — GitHub — Upwork

Profile Summary

Robotics Software Engineer with expertise in autonomous mobile robots, SLAM, navigation, and perception systems. Strong experience with ROS1/ROS2, C++, Linux development environments, and simulation tools like Isaac Sim and Gazebo. Skilled in hardware and software integration for robotics, including experience with RealSense and ZED cameras, motion planning, odometry, and control. Proven ability to solve complex problems independently and deliver robust solutions for robotics applications.

Professional Experience

Robotics Software Engineer (Full-Time)

Wakeb Data Company, Cairo, Egypt — Jan 2025 - Present

- Leading the development and integration of robotic systems with cloud-based platforms using communication protocols like MQTT for scalable data exchange and monitoring.
- Designing and implementing robust robot localization solutions based on wheel odometry, multiple IMUs, and GNSS systems including RTK GPS for high-accuracy outdoor navigation.
- Working on autonomous navigation systems using SLAM and sensor fusion, enabling reliable operation in both indoor and outdoor environments, including GPS-denied areas.
- Implementing untrained navigation using the Mesh Navigation Stack (MeshNav), with custom ROS 2 plugins to handle path planning, terrain adaptation, and non-grid-based environments.
- Collaborating with hardware and software teams to deploy and test robotic solutions across real-world use cases in industrial and logistics sectors.
- Enhancing system reliability by fusing sensor data from GPS, IMU, and encoders using EKF-based sensor fusion techniques within the ROS 2 framework.

Freelance Robotics Software Engineer — Computer Vision Engineer

Upwork — Jan 2023 - Present

- Developed and deployed robotics software solutions for mobile robots, including SLAM (Cartographer, SLAM Toolbox, GMapping) and visual odometry-based mapping for restaurant service robots.
- Extensive experience in odometry, control, and custom SLAM-based navigation systems, including building Nav2 plugins (e.g., Pure Pursuit with ZeroCBF filter).
- Proficient in ROS1/ROS2, MoveIt2, Isaac Sim, Linux development, and high-fidelity simulations using Gazebo.
- Worked with RealSense and ZED cameras for 3D perception, depth sensing, and point cloud analysis in autonomous robotic systems.
- Created object detection models using YOLO and SSD algorithms, with camera calibration via OpenCV.
- Expertise in photogrammetry, point cloud processing using PCL, and 3D data analysis for autonomous robots.
- Designed robotic hardware models in SolidWorks and Autodesk Inventor.

Computer Vision Engineer (Part-Time, Remote)

OXO Technology, Madrid, Spain — Feb 2023 - Present

- Developed advanced detection algorithms and stereo vision techniques to enhance object detection by creating and processing point clouds.

- Worked on color detection algorithms tailored for medical applications, improving diagnostic accuracy.
- Applied 3D data processing techniques using PCL and Open3D to enhance spatial data analysis in medical imaging.

Education

B.Sc. Mechatronics Engineering

Faculty of Engineering, Ain Shams University, Cairo, Egypt

Sep 2018 - Sep 2023 — GPA: 3.2

Graduation Project: *Autonomous Collaborative Assembly Robot using serial-manipulator (IRB 120)* -

Grade: A+

Technical Skills

- **Programming Languages:** C++, Python, C, Bash (automation scripts)
- **Robotics Frameworks:** ROS1, ROS2, MoveIt2, Gazebo, Isaac Sim, MQTT integration
- **Mobile Robotics:** SLAM (Cartographer, SLAM Toolbox, GMapping, LIO-SAM, LOAM, KISS-ICP, MeshSLAM), Odometry, Perception, Path Planning, Mesh Navigation Stack with custom plugins
- **Sensor Fusion & State Estimation:** Extended Kalman Filter (EKF), Unscented Kalman Filter (UKF), factor-graph optimization (iSAM2), graph-based SLAM
- **Robot Arms and Manipulation:** xArm, Inspire Hand (custom ROS2 controllers and MoveIt2 integration), ABB IRB series, Universal Robots (UR5/UR10) for industrial and medical robotics solutions
- **Simulation Tools:** Isaac Sim, Gazebo, MATLAB, Simulink
- **Cameras:** RealSense, ZED, OpenCV for image processing, camera calibration
- **Computer Vision:** YOLO, SSD, Deep SORT, TensorFlow, PyTorch
- **3D Data Processing:** PCL, Open3D, Photogrammetry, Point Cloud Processing
- **Tools:** Linux, Bash scripting, SolidWorks, Autodesk Inventor
- **Version Control:** Git

Key Projects

Autonomous Collaborative Assembly Robot using Serial-Manipulator (IRB 120)

- Designed and developed an autonomous collaborative assembly robot using computer vision for mechanical part detection and manipulation.
- Implemented safety measures for human-robot interaction, ensuring the system could operate safely in a collaborative workspace.
- Utilized advanced control algorithms for precise pick-and-place operations, optimizing the robot's efficiency and accuracy in the assembly process.
- Integrated reverse engineering principles to improve the adaptability of the robot for various assembly tasks.
- Currently in the process of publishing a research paper detailing the system's design, development, and results.

Autonomous Restaurant Delivery Robot

- Designed and developed a fully autonomous robot for restaurant delivery, capable of navigating both indoor and outdoor environments autonomously.
- Employed BLDC hoverboard motors connected to an X11H motor driver, interfaced with an ESP32 microcontroller and integrated with a Raspberry Pi for overall control.
- Implemented LiDAR-based mapping and localization using ROS2, allowing the robot to autonomously map its environment while being controlled via a keyboard interface.

- Developed a ROS2 control system for autonomous navigation, enabling the robot to avoid dynamic obstacles, including tables and chairs, and deliver orders accurately.
- Integrated a GPIO-controlled button on the Raspberry Pi for user intervention, allowing the robot's autonomous operations to be paused and resumed based on button input, with a 30-second timeout for restarting movement.
- Provided full project documentation, including source code, detailed setup instructions, and integration guidelines for the hardware components.
- Collaborated with software and hardware teams to ensure seamless integration of the robot's control system with the restaurant's delivery management software.

Advanced Sensor Fusion for Autonomous Vehicle Platform

- **Client:** Confidential AV Startup (8 months)
- Designed and implemented a real-time sensor-fusion pipeline in ROS 2 combining Velodyne VLP-32C LiDAR, stereo RGB cameras, MEMS IMU, and RTK-GPS.
- Developed an Extended Kalman Filter + sliding-window bundle adjustment backend for sub-5 cm pose drift over 5 km urban test routes.
- Integrated with Autoware Auto, tuning perception and localization modules; validated in closed-course trials.

ROS 2 & Nav2-Based Urban Delivery Robot

- **Client:** Wakeb Data Company (6 months)
- Engineered a last-mile delivery stack on a Clearpath Husky, using Nav2 for waypoint navigation, dynamic obstacle avoidance, and recovery behaviors.
- Implemented NDT-based LiDAR localization with live occupancy-grid mapping at 10 Hz.
- Achieved reliable route completion in dense pedestrian zones, reducing intervention events by 75%.

Off-Road AV Simulation in NVIDIA Isaac Sim

- **Research Collaboration (3 months)**
- Built a realistic off-road terrain with elevation maps, unstructured obstacles, and dynamic actors in Isaac Sim.
- Modeled sensors (LiDAR, stereo cameras, GNSS) and bridged them into ROS 2 for full-stack closed-loop simulation.
- Ran batch tests of path-planning and control algorithms, achieving a 90% success rate on unpaved tracks.

Highway Lane-Keeping & Adaptive Cruise Control Module

- **In-house R&D (4 months)**
- Developed vision-based lane detection using YOLOv5 + inverse perspective mapping, fused with high-definition map waypoints.
- Designed an MPC-based adaptive cruise controller fusing radar and camera data to maintain safe following distances.
- Validated performance on HIL rigs and recorded datasets, aligning with ISO 26262 safety targets.

“Smart Surgical Assistant” Integration

- **Client:** Hypercube Motors (Turnkey Hospital Robotics)
- Integrated the xArm6 robotic arm with the Inspire Hand gripper via Modbus RTU, enabling precise grasp-and-release motions for surgical tool handling.
- Developed a Leap Motion-based teleoperation interface to track surgeon hand movements with deadband and low-pass filtering for smooth robotic control.
- Unified the arm/gripper system with a mobile base to create a fully autonomous “Smart Surgical Assistant” capable of navigating hospital corridors under sterile-field protocols.
- Implemented secure AES-GCM socket communication between the teleoperation station and mobile robot to safeguard patient data and control commands.
- Validated sub-millimeter positioning accuracy and reliable obstacle avoidance in both simulated and real-world hospital floor trials.

Certifications

- Self-Driving Car Engineer Nanodegree, Udacity (2024)
- Sensor Fusion Engineer Nanodegree, Udacity (2024)
- Robotics Software Engineer Nanodegree, Udacity (2023)
- Deep Learning Specialization, DeepLearning.AI (2023)
- Supervised Machine Learning: Regression and Classification, Stanford Online (2022)

Languages

- **English:** Fluent
- **Arabic:** Native